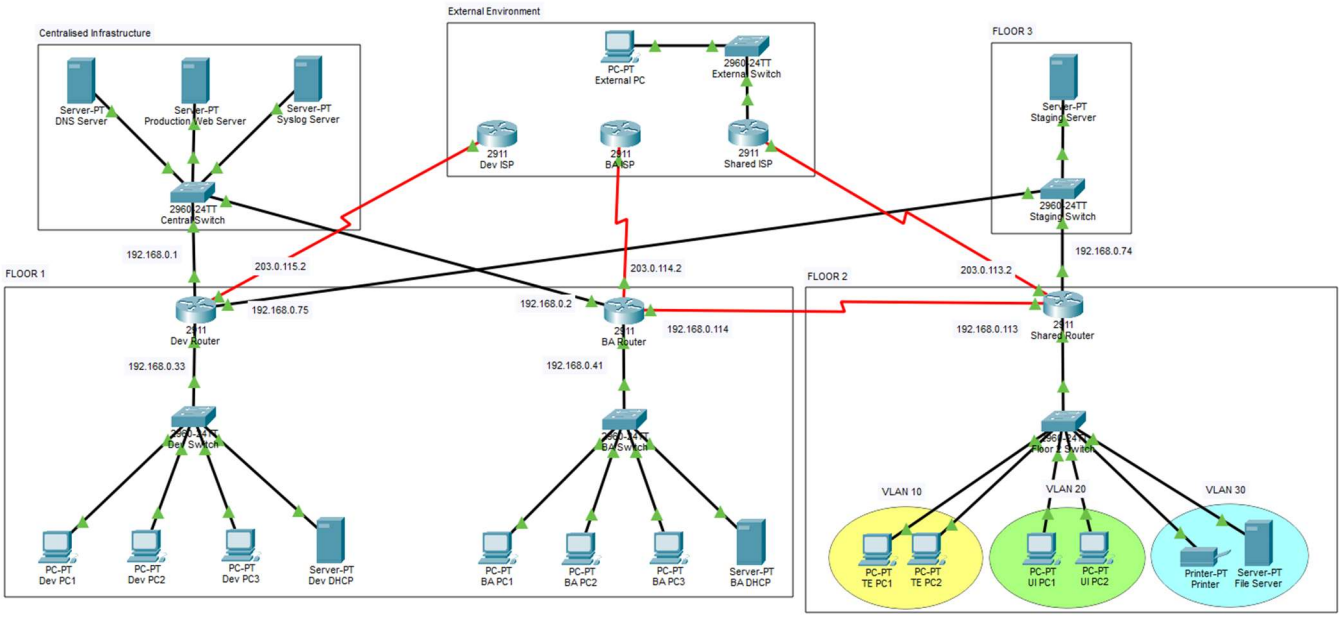
BrightSoft Solutions Network Topology Overview



Floor 1

- Consists of the Developers and Business Analyst Subnet. Each subnet has its own dedicated DHCP server to automatically assign Ip addresses for the 3 end devices. Both subnets have a layer 2 switch that connects to their own Router which connects them to the centralised Infrastructure and ISPs.

Floor 2

- Consists of the Test Engineers, UI Designers and shared recourses subnets. All subnets share the same floor but are logically separated via inter-VLAN routing. All are connected to the same layer 2 switch which connects them to the Shared router.

Floor 3

- Consists of the Staging Web server which is connected to a layer 2 switch that connects it to the Shared router.

Centralised Infrastructure

- Consists of DNS, Production Web and Syslog Server which is accessible to all 3 floors.

External Environment

- Consists of Developers, Business Analysts and Shared ISPs. This simulates the internet and an external pc is used to simulate External access.

IP Subnet Scheme for BrightSoft Solutions Network Topology

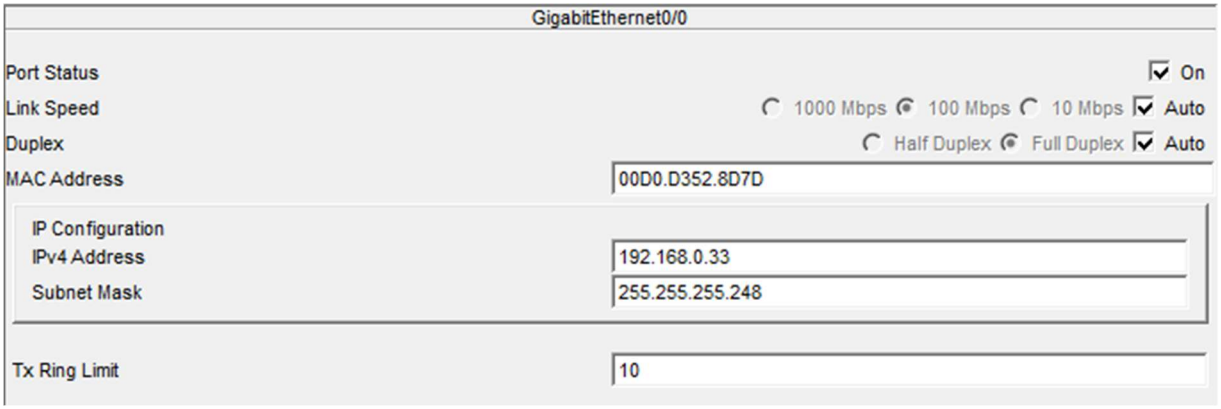
Subnet	Subnet Network Address	Subnet Mask	Subnet Usable Range	Used for
Central Lan	192.168.0.0/27	255.255.255.224	192.168.0.1 - 192.168.0.30	Central Infrastructure and

				Router Connections
Developers	192.168.0.32/29	255.255.255.248	192.168.0.33 - 192.168.0.38	Developers
Business Analysts	192.168.0.40/29	255.255.255.248	192.168.0.41 - 192.168.0.46	Business Analysts
Test Engineers VLAN 10	192.168.0.48/29	255.255.255.248	192.168.0.49 - 192.168.0.54	Test Engineers
UI Designers VLAN 20	192.168.0.56/29	255.255.255.248	192.168.0.57 - 192.168.0.62	UI Designers
Shared Resources VLAN 30	192.168.0.64/29	255.255.255.248	192.168.0.65 - 192.168.0.70	Printer and file server
Staging	192.168.0.72/29	255.255.255.248	192.168.0.73 - 192.168.0.78	Staging web server
Backup Link	192.168.0.112/30	255.255.255.252	192.168.0.113 - 192.168.0.114	Backup link for shared router and BA router
Dev ISP	203.0.115.0/30	255.255.255.252	203.0.115.1 - 203.0.115.2	Internet
BA ISP	203.0.114.0/30	255.255.255.252	203.0.114.1 - 203.0.114.2	Internet
Shared ISP	203.0.113.0/30	255.255.255.252	203.0.113.1 - 203.0.113.2	Internet
External PC	192.168.1.0/24	255.255.255.0	192.168.1.1 - 192.168.1.254	Testing External Access

(Calculator.net, 2026)

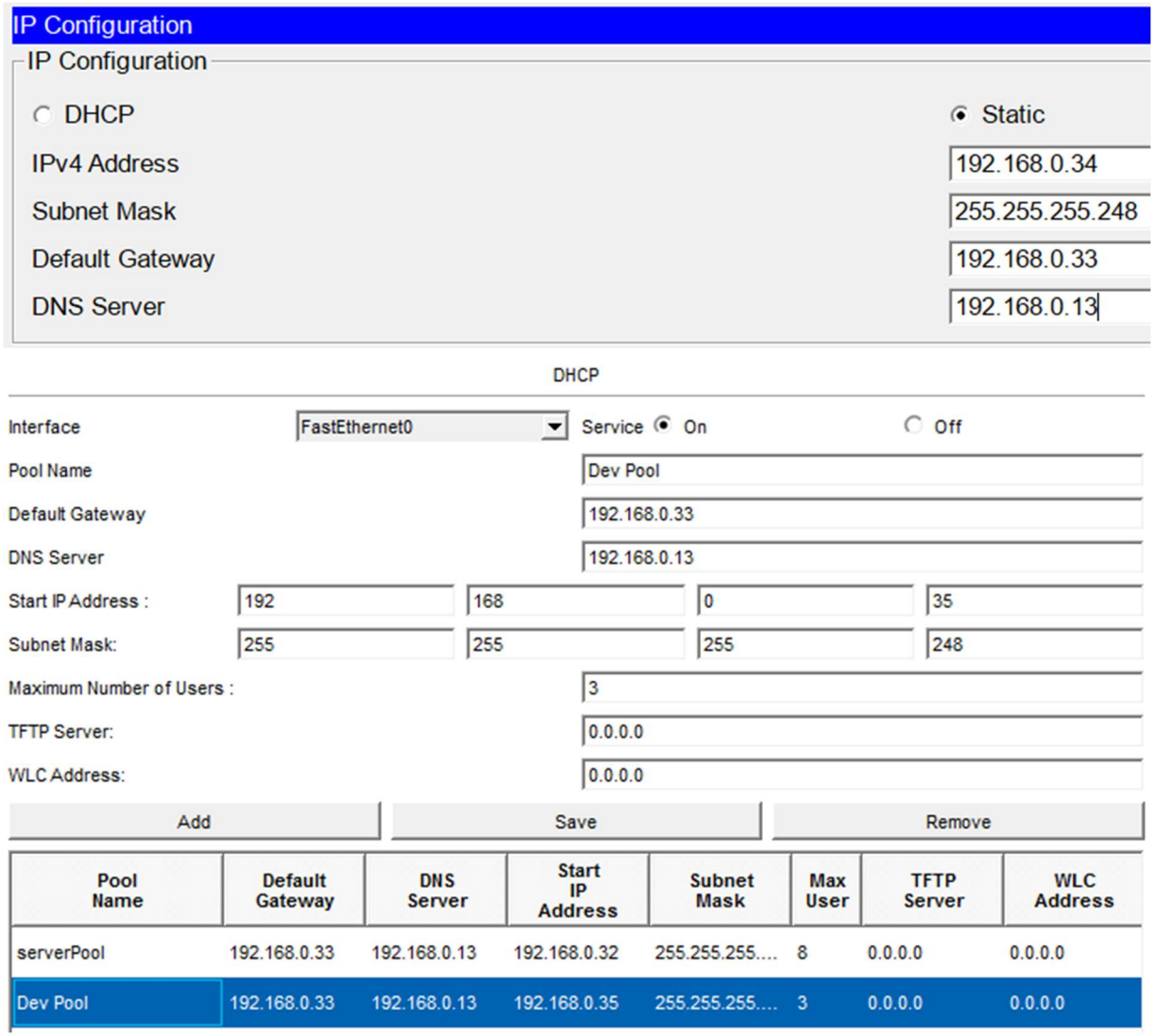
Developer Team Subnet

Dev Router Port Configuration



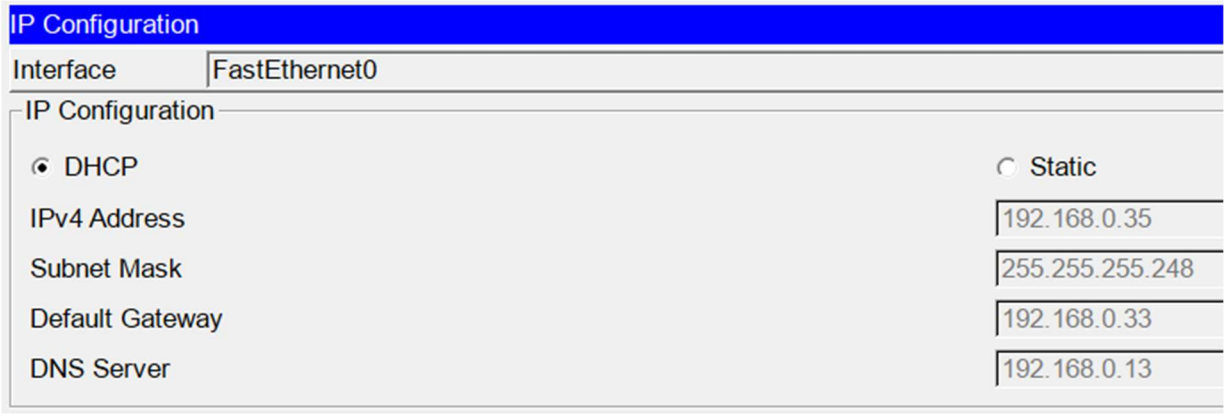
(DHCP Server Configuration Tutorial With Multiple Switches using CISCO Packet Tracer, 2023)

Dev DHCP IP and DHCP Pool Configuration



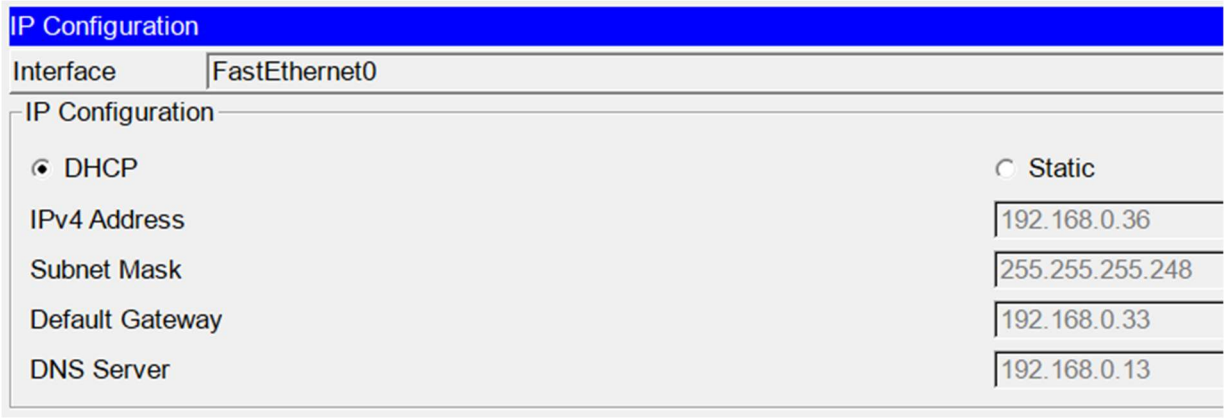
(DHCP Server Configuration Tutorial With Multiple Switches using CISCO Packet Tracer, 2023)

Dev PC 1 IP Configuration



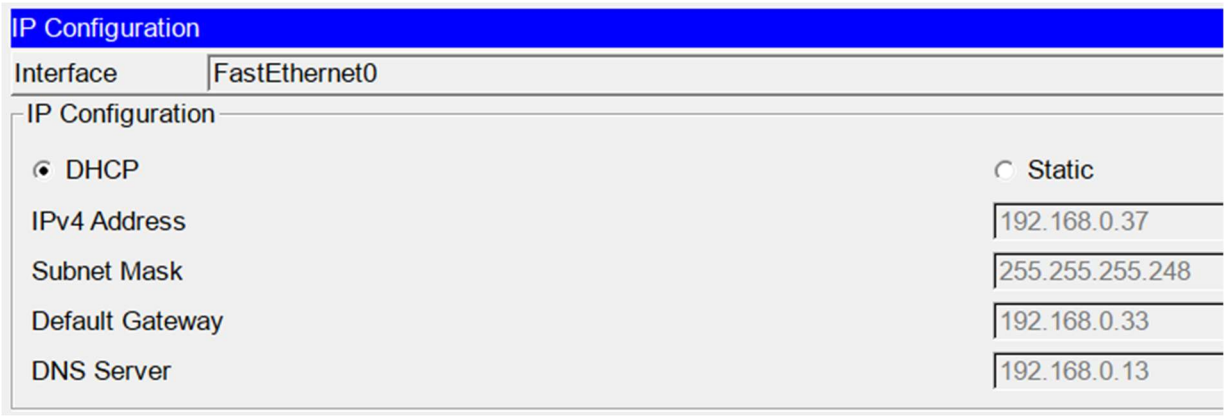
(DHCP Server Configuration Tutorial With Multiple Switches using CISCO Packet Tracer, 2023)

Dev PC 2 IP Configuration



(DHCP Server Configuration Tutorial With Multiple Switches using CISCO Packet Tracer, 2023)

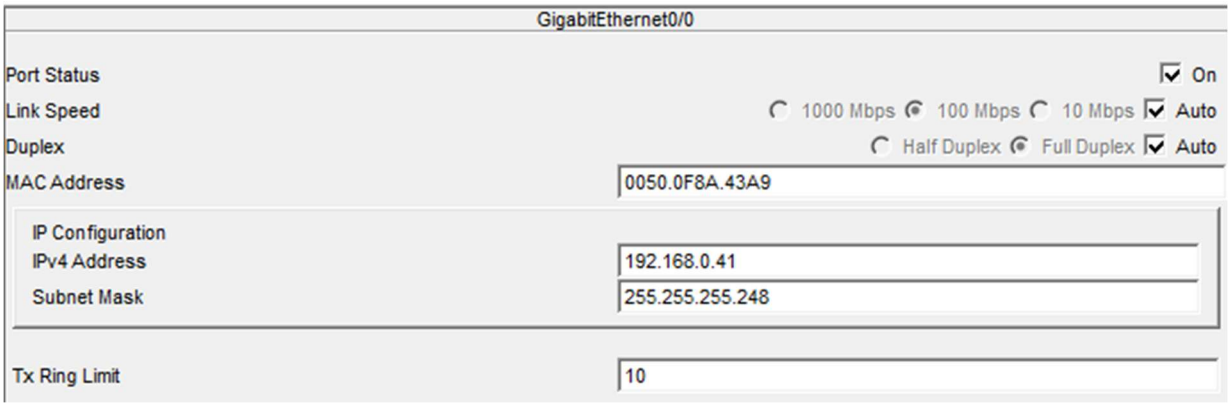
Dev PC 3 IP Configuration



(DHCP Server Configuration Tutorial With Multiple Switches using CISCO Packet Tracer, 2023)

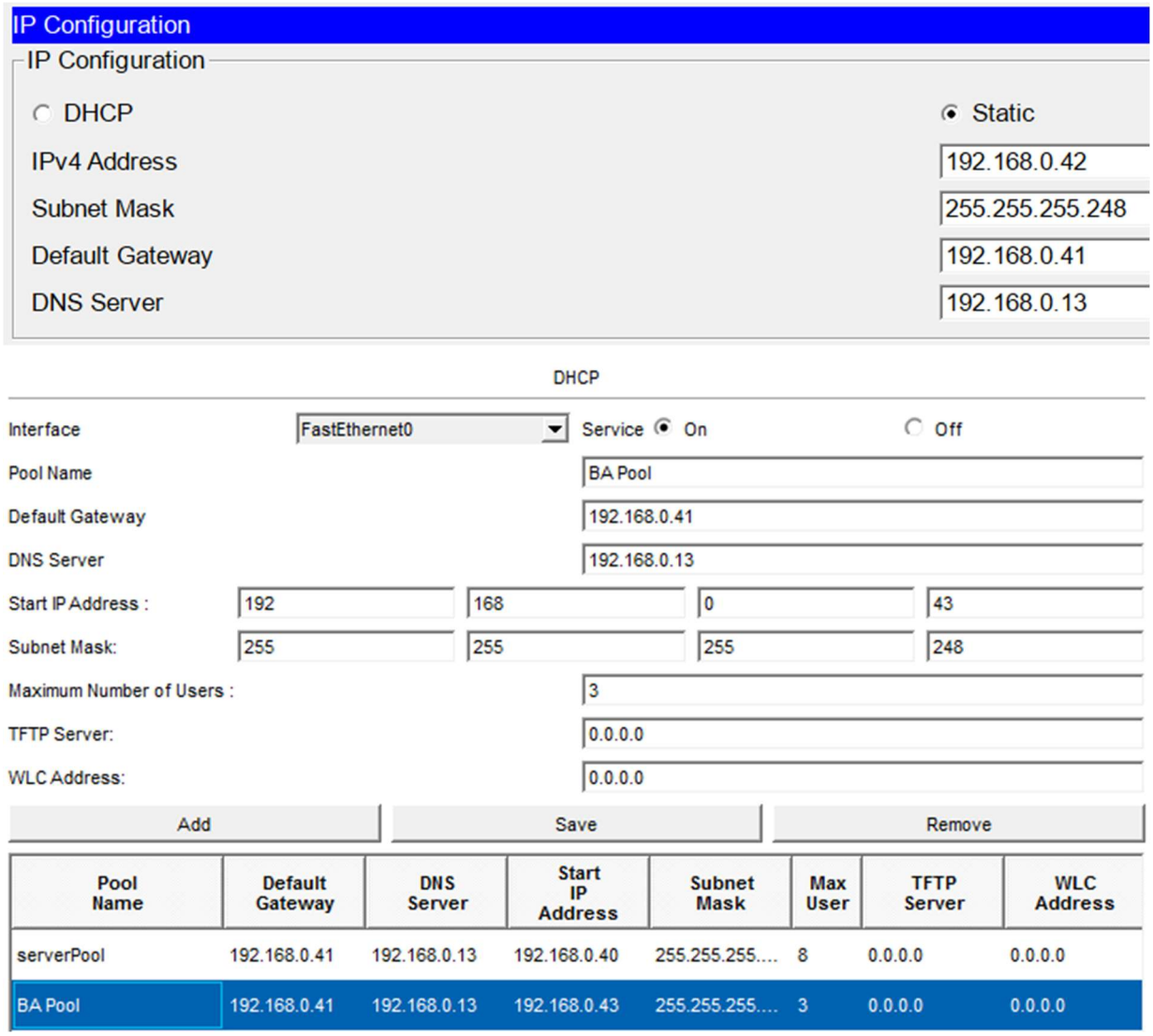
Business Analysts Team Subnet

BA Router Port Configuration



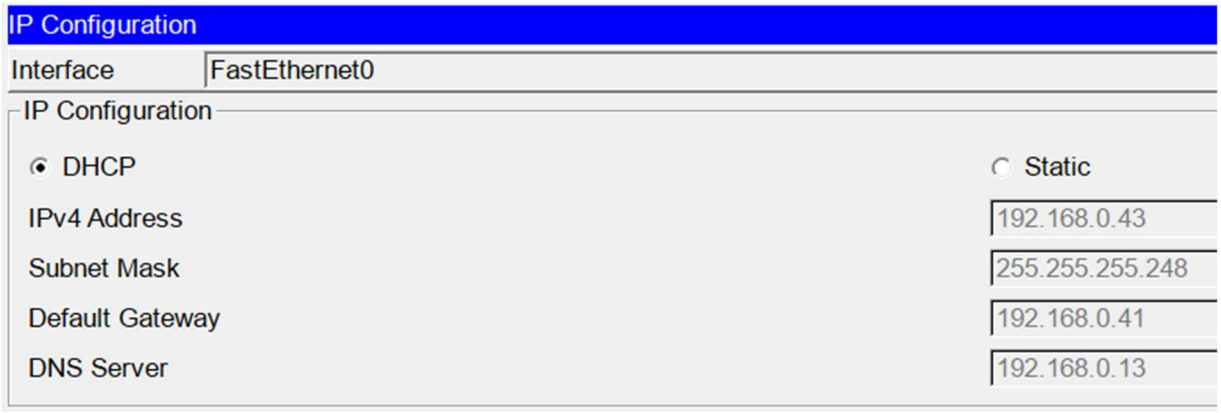
(DHCP Server Configuration Tutorial With Multiple Switches using CISCO Packet Tracer, 2023)

BA DHCP Server IP and DHCP Configuration



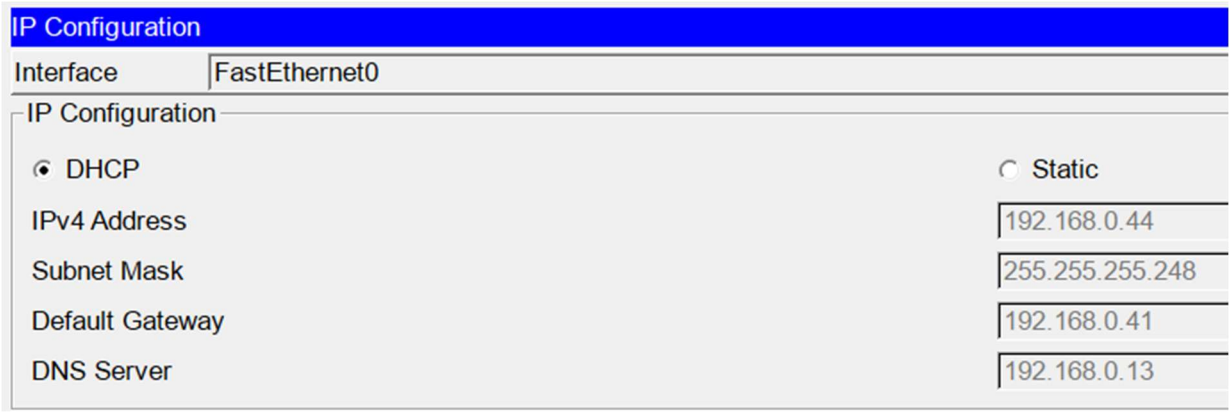
(DHCP Server Configuration Tutorial With Multiple Switches using CISCO Packet Tracer, 2023)

BA PC 1 IP Configuration



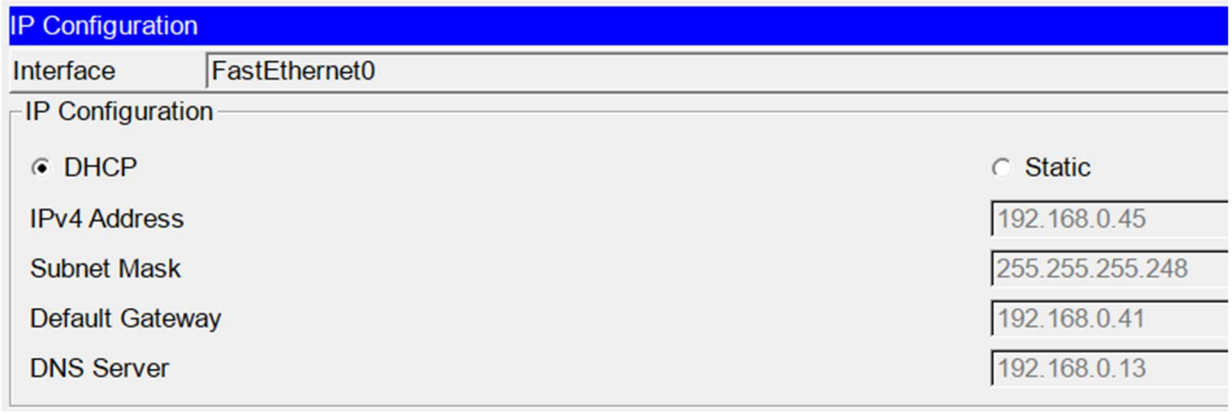
(DHCP Server Configuration Tutorial With Multiple Switches using CISCO Packet Tracer, 2023)

BA PC 2 IP Configuration



(DHCP Server Configuration Tutorial With Multiple Switches using CISCO Packet Tracer, 2023)

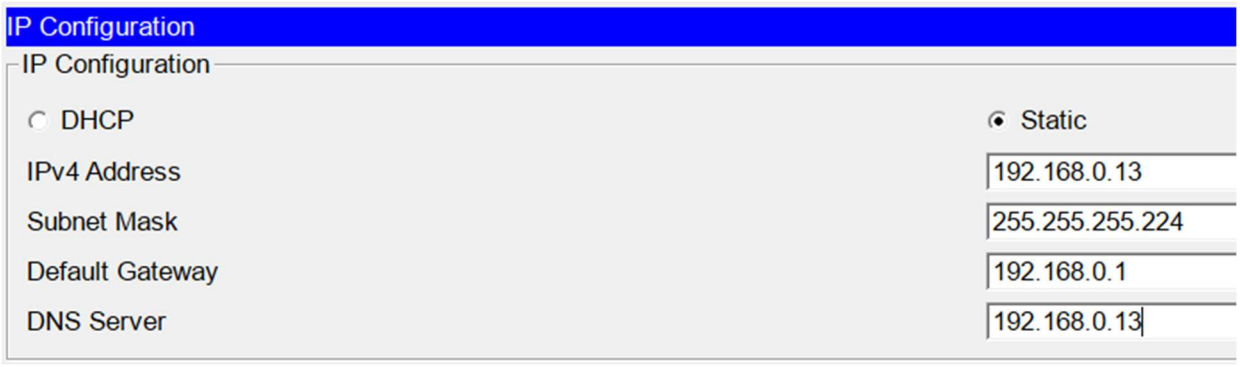
BA PC 3 IP Configuration



(DHCP Server Configuration Tutorial With Multiple Switches using CISCO Packet Tracer, 2023)

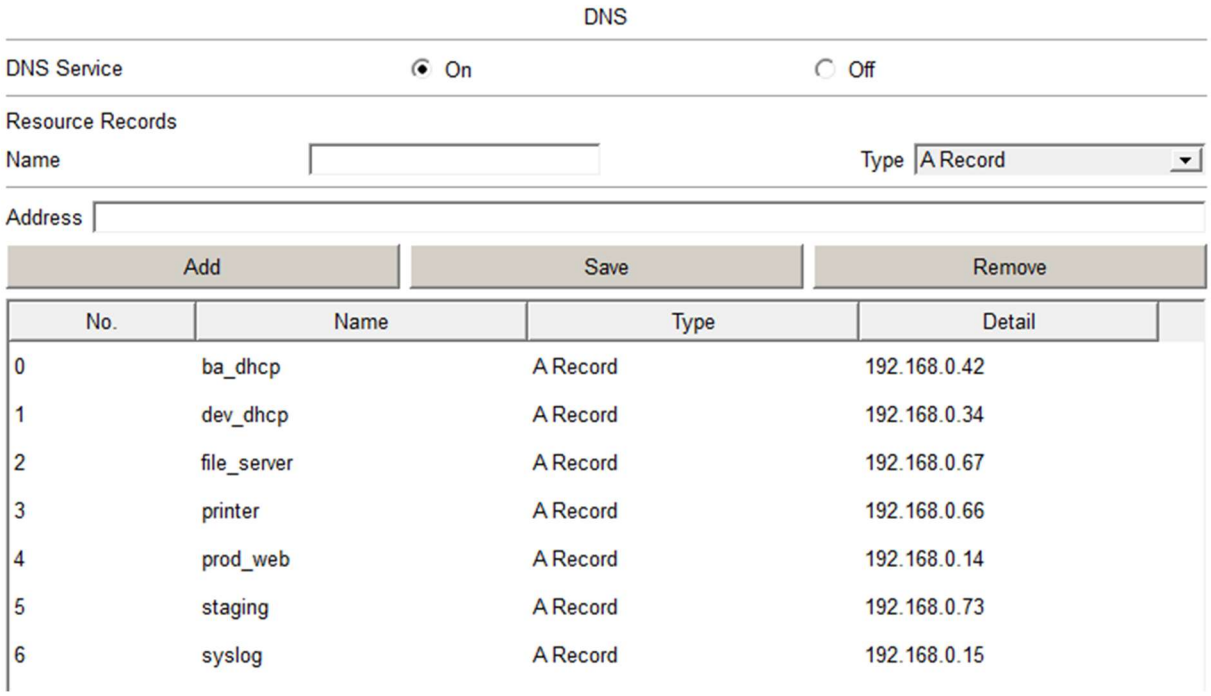
DNS Configuration

DNS IP Configuration



(How to configure DNS Server | Step by step DNS server configuration, 2024)

DNS Service Configuration to allow resolving hostnames for internal devices across the organization.



No.	Name	Type	Detail
0	ba_dhcp	A Record	192.168.0.42
1	dev_dhcp	A Record	192.168.0.34
2	file_server	A Record	192.168.0.67
3	printer	A Record	192.168.0.66
4	prod_web	A Record	192.168.0.14
5	staging	A Record	192.168.0.73
6	syslog	A Record	192.168.0.15

(How to configure DNS Server | Step by step DNS server configuration, 2024)

VLAN Configuration on Floor 2 Switch

VLAN creation

```
Floor_2_Switch(config)#vlan 10
Floor_2_Switch(config-vlan)#name Test_Engineers
Floor_2_Switch(config-vlan)#ex
Floor_2_Switch(config)#
Floor_2_Switch(config)#vlan 20
Floor_2_Switch(config-vlan)#name UI_Designers
Floor_2_Switch(config-vlan)#ex
Floor_2_Switch(config)#
Floor_2_Switch(config)#vlan 30
Floor_2_Switch(config-vlan)#name Shared_Resources
Floor_2_Switch(config-vlan)#ex
```

(CCNA DAY 24: Configure Router-on-a-Stick Inter-VLAN Routing | Inter-VLAN Routing Router on a Stick, 2023)

VLAN port assignment

```
Floor_2_Switch(config)#interface range fa0/1-2
Floor_2_Switch(config-if-range)#switchport mode access
Floor_2_Switch(config-if-range)#switchport access vlan 10
Floor_2_Switch(config-if-range)#ex
Floor_2_Switch(config)#
Floor_2_Switch(config)#interface range fa0/3-4
Floor_2_Switch(config-if-range)#switchport mode access
Floor_2_Switch(config-if-range)#switchport access vlan 20
Floor_2_Switch(config-if-range)#ex
Floor_2_Switch(config)#
Floor_2_Switch(config)#interface range fa0/5-6
Floor_2_Switch(config-if-range)#switchport mode access
Floor_2_Switch(config-if-range)#switchport access vlan 30
Floor_2_Switch(config-if-range)#ex
```

(CCNA DAY 24: Configure Router-on-a-Stick Inter-VLAN Routing | Inter-VLAN Routing Router on a Stick, 2023)

Trunk configuration to shared router

```
Floor_2_Switch(config)#interface gig0/1
Floor_2_Switch(config-if)#switchport mode trunk

Floor_2_Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to down

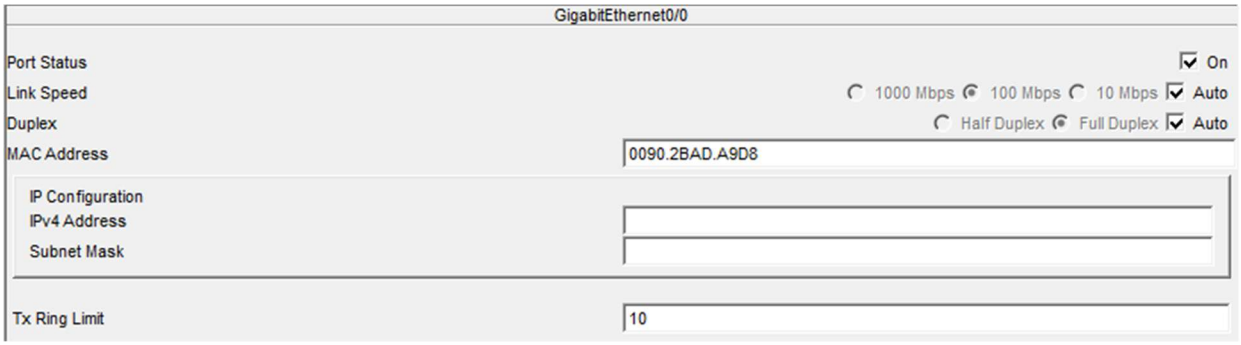
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to up

Floor_2_Switch(config-if)#switchport trunk allowed vlan 10,20,30
Floor_2_Switch(config-if)#ex
```

(CCNA DAY 24: Configure Router-on-a-Stick Inter-VLAN Routing | Inter-VLAN Routing Router on a Stick, 2023)

Router-on-a-Stick – Ip routing protocol

Port on shared router to the floor 2 subnet



(CCNA DAY 24: Configure Router-on-a-Stick Inter-VLAN Routing | Inter-VLAN Routing Router on a Stick, 2023)

Shared router sub-interface configuration

```
Shared_Router(config)#interface gig0/0.10
Shared_Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed
state to up

Shared_Router(config-subif)#encapsulation dot1Q 10
Shared_Router(config-subif)#ip address 192.168.0.49 255.255.255.248
Shared_Router(config-subif)#ex
Shared_Router(config)#
Shared_Router(config)#interface gig0/0.20
Shared_Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed
state to up

Shared_Router(config-subif)#encapsulation dot1Q 20
Shared_Router(config-subif)#ip address 192.168.0.57 255.255.255.248
Shared_Router(config-subif)#ex
Shared_Router(config)#
Shared_Router(config)#interface gig0/0.30
Shared_Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed
state to up

Shared_Router(config-subif)#encapsulation dot1Q 30
Shared_Router(config-subif)#ip address 192.168.0.65 255.255.255.248
Shared_Router(config-subif)#ex
```

(CCNA DAY 24: Configure Router-on-a-Stick Inter-VLAN Routing | Inter-VLAN Routing Router on a Stick, 2023)

VLAN 10 ACL Configuration on the Shared Router

ACL created to deny VLAN 20 access and Stagging web server as well as permit DNS, Shared resources access.

```
Shared_Router(config)#ip access-list extended Block_Traffic_Vlan10
Shared_Router(config-ext-nacl)#permit ip 192.168.0.48 0.0.0.7 192.168.0.64 0.0.0.7
Shared_Router(config-ext-nacl)#permit ip 192.168.0.48 0.0.0.7 host 192.168.0.13
Shared_Router(config-ext-nacl)#deny ip 192.168.0.48 0.0.0.7 192.168.0.56 0.0.0.7
Shared_Router(config-ext-nacl)#deny ip 192.168.0.48 0.0.0.7 host 192.168.0.73
Shared_Router(config-ext-nacl)#deny ip any any
Shared_Router(config-ext-nacl)#ex
Shared_Router(config)#interface gig0/0.10
Shared_Router(config-subif)#ip access-group Block_Traffic_Vlan10 in
Shared_Router(config-subif)#ex
```

(CCNA DAY 43: Extended ACL Configuration Packet Tracer| How to Configure Extended Access Control List, 2023)

VLAN 20 ACL Configuration on the Shared Router

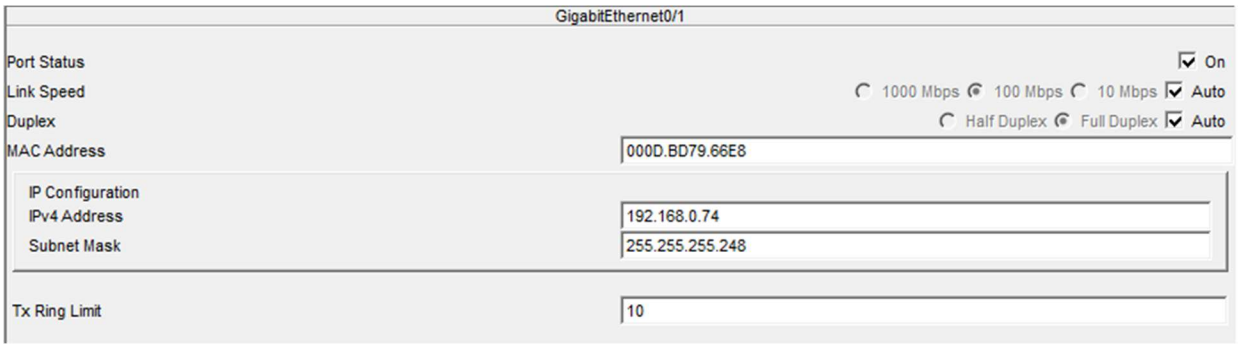
ACL created to deny VLAN 10 access and Stagging web server as well as permit DNS, Shared resources access.

```
Shared_Router(config)#ip access-list extended Block_Traffic_Vlan20
Shared_Router(config-ext-nacl)#permit ip 192.168.0.56 0.0.0.7 192.168.0.64 0.0.0.7
Shared_Router(config-ext-nacl)#permit ip 192.168.0.56 0.0.0.7 host 192.168.0.13
Shared_Router(config-ext-nacl)#deny ip 192.168.0.56 0.0.0.7 192.168.0.48 0.0.0.7
Shared_Router(config-ext-nacl)#deny ip 192.168.0.56 0.0.0.7 host 192.168.0.73
Shared_Router(config-ext-nacl)#deny ip any any
Shared_Router(config-ext-nacl)#ex
Shared_Router(config)#interface gig0/0.20
Shared_Router(config-subif)#ip access-group Block_Traffic_Vlan20 in
Shared_Router(config-subif)#ex
```

(CCNA DAY 43: Extended ACL Configuration Packet Tracer| How to Configure Extended Access Control List, 2023)

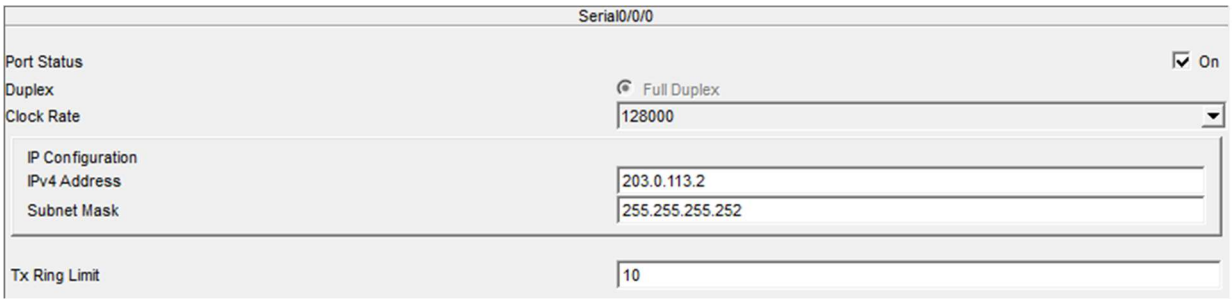
NAT Configurations

- Shared router Staging default gateway configuration



(Static NAT Configuration | Network Address Translation | Cisco Packet Tracer, 2024)

- Shared router Isp Ip configuration



(Static NAT Configuration | Network Address Translation | Cisco Packet Tracer, 2024)

- Nat configuration to Translate Public IP to Private IP Address on the Shared router

```
Shared_Router(config)#ip nat inside source static tcp 192.168.0.73 80
203.0.113.2 80
Shared_Router(config)#ip nat inside source static tcp 192.168.0.73 443
203.0.113.2 443
Shared_Router(config)#
Shared_Router(config)#
Shared_Router(config)#interface gig0/1
Shared_Router(config-if)#ip nat inside
Shared_Router(config-if)#ex
Shared_Router(config)#
Shared_Router(config)#interface serial0/0/0
Shared_Router(config-if)#ip nat outside
Shared_Router(config-if)#ex
```

(Static NAT Configuration | Network Address Translation | Cisco Packet Tracer, 2024)

Extended Access Lists

- Acl to block internal access to staging but allow external environment to access staging http/https

```
Shared_Router(config)#ip access-list extended External_Access
Shared_Router(config-ext-nacl)#permit tcp any host 203.0.113.2 eq 80
Shared_Router(config-ext-nacl)#permit tcp any host 203.0.113.2 eq 443
Shared_Router(config-ext-nacl)#deny ip any any
Shared_Router(config-ext-nacl)#ex
Shared_Router(config)#
Shared_Router(config)#interface serial0/0/0
Shared_Router(config-if)#ip access-group External_Access in
Shared_Router(config-if)#ex
```

(CCNA DAY 43: Extended ACL Configuration Packet Tracer| How to Configure Extended Access Control List, 2023)

- Block all traffic except dev subnet

```
Shared_Router(config)#ip access-list extended Internal_Access
Shared_Router(config-ext-nacl)#permit tcp any host 192.168.0.73 eq 80
Shared_Router(config-ext-nacl)#permit tcp any host 192.168.0.73 eq 443
Shared_Router(config-ext-nacl)#permit ip 192.168.0.32 0.0.0.7 host 192.168.0.73
Shared_Router(config-ext-nacl)#deny ip any host 192.168.0.73
Shared_Router(config-ext-nacl)#permit ip any any
Shared_Router(config-ext-nacl)#ex
Shared_Router(config)#
Shared_Router(config)#interface gig0/1
Shared_Router(config-if)#ip access-group Internal_Access out
Shared_Router(config-if)#ex
```

(CCNA DAY 43: Extended ACL Configuration Packet Tracer| How to Configure Extended Access Control List, 2023)

- Manage staging traffic

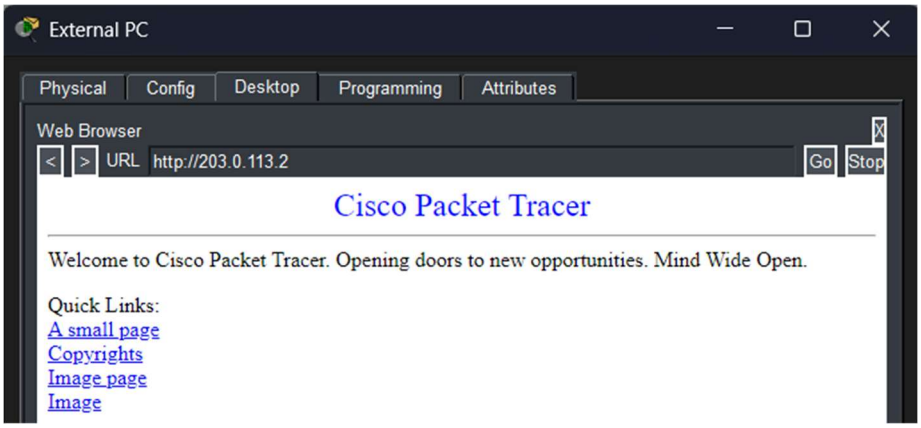
```
Shared_Router(config)#ip access-list extended From_Staging
Shared_Router(config-ext-nacl)#permit icmp host 192.168.0.73 any
Shared_Router(config-ext-nacl)#permit tcp host 192.168.0.73 any established
Shared_Router(config-ext-nacl)#deny ip host 192.168.0.73 any
Shared_Router(config-ext-nacl)#permit ip any any
Shared_Router(config-ext-nacl)#ex
Shared_Router(config)#
Shared_Router(config)#interface gig0/1
Shared_Router(config-if)#ip access-group From_Staging in
Shared Router(config-if)#ex
```

(CCNA DAY 43: Extended ACL Configuration Packet Tracer| How to Configure Extended Access Control List, 2023)

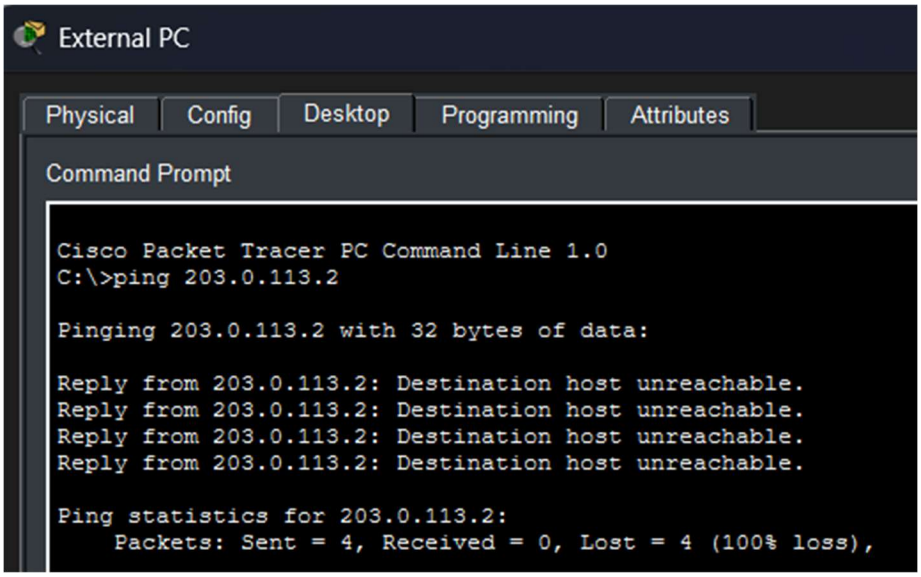
Screenshot Evidence for Configuration

External Accessability to Staging Web Server

1. External Access to Staging Web Server Webpage



2. External Ping to Staging Web Server



3. IP NAT Translations and External ACL Hit Counts

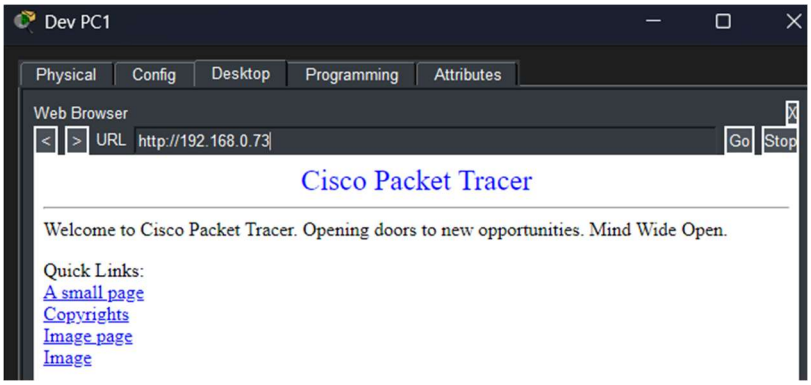
```
Shared_Router>enable
Shared_Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
tcp 203.0.113.2:443    192.168.0.73:443   ---                ---
tcp 203.0.113.2:80     192.168.0.73:80    ---                ---
tcp 203.0.113.2:80     192.168.0.73:80    192.168.1.10:1025
192.168.1.10:1025

Shared_Router#show access-lists External_Access
Extended IP access list External_Access
    permit tcp any host 203.0.113.2 eq www (6 match(es))
    permit tcp any host 203.0.113.2 eq 443
    deny ip any any (4 match(es))
```

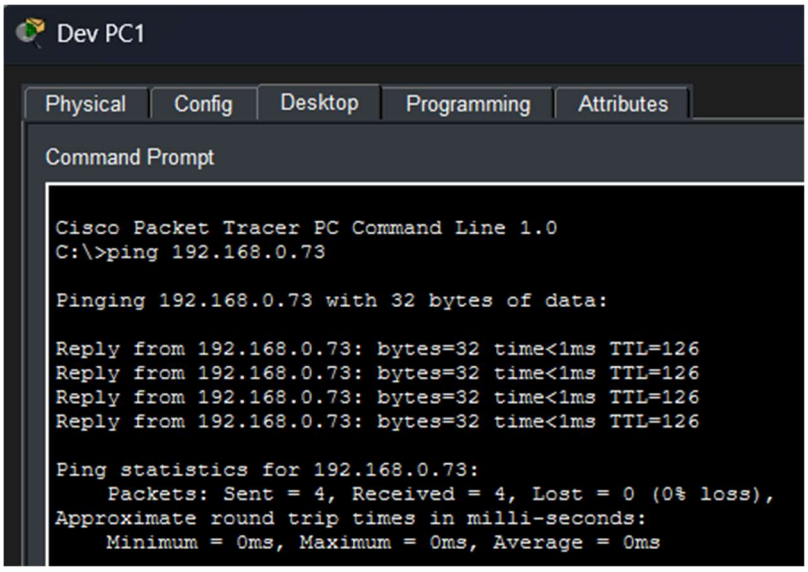
As shown by the Screenshots external partners are permitted access via TCP ports 80 and 443 known as HTTP/HTTPS only and denied all other protocols.

Internal Accessability to Staging Web Server

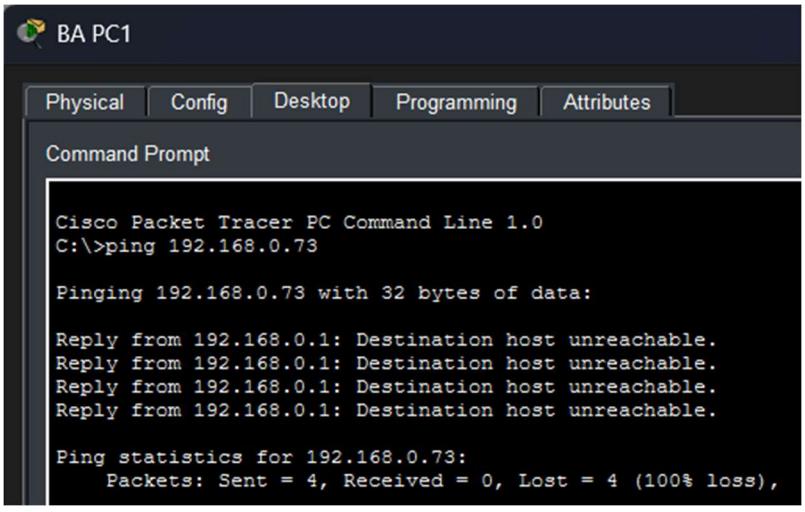
1. Developers Team Accessing Staging Web Server webpage



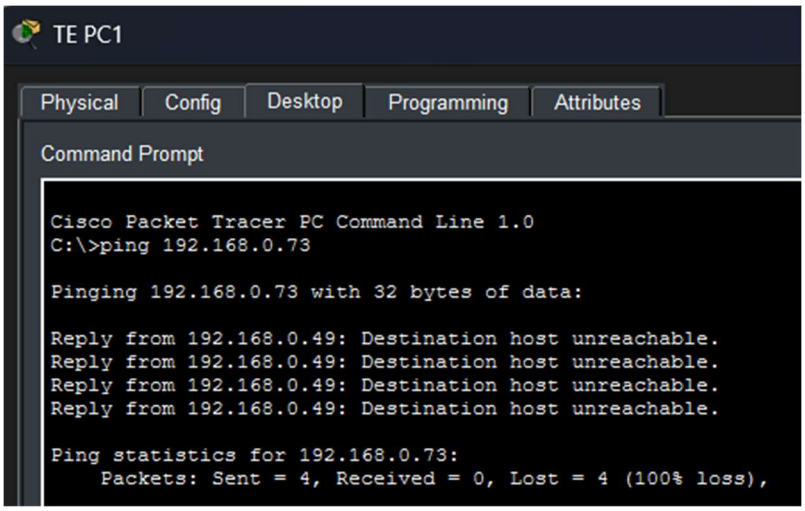
2. Developers Team ping to Staging Web Server



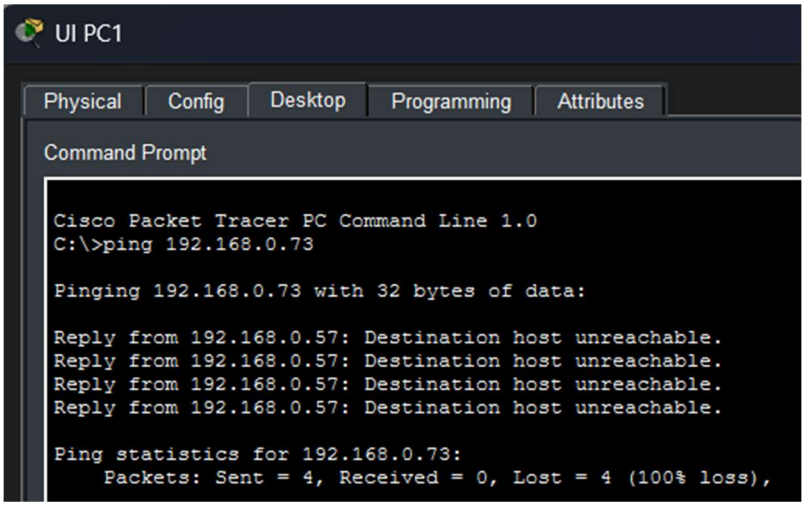
3. Business Analyst Team ping to Staging Web Server



4. Test Engineers ping to Staging Web Server



5. UI Designers ping to Staging Web Server



6. Internal ACL Hit Counts

```
Shared_Router#show access-lists Internal_Access
Extended IP access list Internal_Access
    permit tcp any host 192.168.0.73 eq www (6 match(es))
    permit tcp any host 192.168.0.73 eq 443
    permit ip 192.168.0.32 0.0.0.7 host 192.168.0.73
    deny ip any host 192.168.0.73
    permit ip any any (10 match(es))
```

As shown by the screenshots the developer team are the only team that can access the staging web server via ping and webpage which enables the team to update the staging web server.

For the BrightSoft theory network design, the staging server needs to be placed in a particular DMZ zone known as the Staging VLAN, which is defined as 192.168.0.72/29. This ensures a buffer zone between the Internet and our internal network, so that our external partners who require access to our client demos do not have access to our internal network, for instance, our production web server, which is 192.168.0.14 (ExamLabs, n.d). The buffer zone ensures a tight security level for our Shared Router, which is the gateway to our staging subnet. For the data plan, we would set extended ACLs on our Shared Router, which would adhere to the principle of least privilege. The ACLs would allow incoming HTTP from port 80 and HTTPS from port 443 from our external partners to our staging server 192.168.0.73, while denying all other external partner traffic. On the internal network, our data plan ACLs would allow our developers group 192.168.0.32/29 to access our staging network for updates or maintenance, while denying access to our Business Analysts, Test Engineers, and UI Designers. For the control plane, we would use OSPF to maintain routing through our internal network. The ISP interface would be set to passive, so our internal topology is not advertised (Awati, 2023).

Supporting services play a crucial role in the functionality as well as the security of this segment. For the DNS resolution service, the central DNS server at 192.168.0.13 would be utilized to resolve the internal requests to the staging environment as well as the external requests from the partners. In the case of the certificate service, the SSL/TLS certificates would be deployed to the staging server to ensure that all the demonstrations to the partners are done through an encrypted channel, protecting client information during the “partially externally accessible” session (elena, 2025). Lastly, with regards to the monitoring service, all the traffic logs and security events occurring in the staging environment would be forwarded to the centralized Syslog server at 192.168.0.15. This design choice would limit the blast radius of a potential security breach to a great extent. In the event the staging server is compromised by an external entity, the ACLs configured on the Shared router would prevent the attacker from moving laterally to the internal Floor 1 or Floor 2 networks. From the operational security viewpoint, the design would cater to the Developers requirement of having the system updated frequently while providing a secure platform for the partner demonstrations. By having all the activities and traffic channeled through the central Syslog server, the operational security would ensure that the environment is secure while adhering to the current networking standards (N-able, 2026).

RIP Version 2 Configuration

- Developer Router

```
Dev_Router(config)#router rip
Dev_Router(config-router)#version 2
Dev_Router(config-router)#network 192.168.0.0
Dev_Router(config-router)#no auto-summary
Dev Router(config-router)#ex
```

(Cisco RIP v2 Setup in 5 Minutes | CISCO Packet Tracer, 2025)

- Business Analysts Router

```
BA_Router(config)#router rip
BA_Router(config-router)#version 2
BA_Router(config-router)#network 192.168.0.0
BA_Router(config-router)#no auto-summary
BA_Router(config-router)#ex
```

(Cisco RIP v2 Setup in 5 Minutes | CISCO Packet Tracer, 2025)

- Shared Router

```
Shared_Router(config)#router rip
Shared_Router(config-router)#version 2
Shared_Router(config-router)#network 192.168.0.0
Shared_Router(config-router)#no auto-summary
Shared_Router(config-router)#passive-interface default
Shared_Router(config-router)#no passive-interface gig0/1
Shared_Router(config-router)#no passive-interface serial0/0/1
Shared_Router(config-router)#ex
```

(Cisco RIP v2 Setup in 5 Minutes | CISCO Packet Tracer, 2025)

OSPF Configuration

- Shared Router

```
Shared_Router(config)#router ospf 1
Shared_Router(config-router)#network 192.168.0.48 0.0.0.7 area 0
Shared_Router(config-router)#network 192.168.0.56 0.0.0.7 area 0
Shared_Router(config-router)#network 192.168.0.64 0.0.0.7 area 0
Shared_Router(config-router)#network 192.168.0.72 0.0.0.7 area 0
```

(Cisco OSPF Configuration Explained: Step-by-Step Guide for Beginners! 🚀 , 2025)

Redistribution

- Shared Router

```
Shared_Router(config)#router rip
Shared_Router(config-router)#redistribute ospf 1 metric 1
Shared_Router(config-router)#ex

Shared_Router(config)#router ospf 1
Shared_Router(config-router)#redistribute rip subnets
Shared_Router(config-router)#ex
```

(Redistribution OSPF with Eigrp Lab in Packe Tracer | Networkforyou | CCNA, 2022)

IP Protocol Configuration Screenshots

1. Developer Router

```
Dev_Router>en
Dev_Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 17 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface             Send  Recv  Triggered RIP  Key-chain
  GigabitEthernet0/0    22
  GigabitEthernet0/1    22
  GigabitEthernet0/2    22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
           192.168.0.0
Passive Interface(s):
Routing Information Sources:
           Gateway         Distance      Last Update
           192.168.0.2          120      00:00:01
           192.168.0.74         120      00:00:15
Distance: (default is 120)
```

2. Business Analysts Router

```
BA_Router>en
BA_Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 23 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface             Send  Recv  Triggered RIP  Key-chain
  GigabitEthernet0/0    22
  GigabitEthernet0/1    22
  Serial0/0/1           22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
          192.168.0.0
Passive Interface(s):
Routing Information Sources:
          Gateway         Distance      Last Update
          192.168.0.1           120      00:00:04
          192.168.0.113         120      00:00:07
Distance: (default is 120)
```

3. Shared Router

```
Shared_Router>en
Shared_Router#show running-config | section router
router ospf 1
 log-adjacency-changes
 redistribute rip subnets
 network 192.168.0.48 0.0.0.7 area 0
 network 192.168.0.56 0.0.0.7 area 0
 network 192.168.0.64 0.0.0.7 area 0
 network 192.168.0.72 0.0.0.7 area 0
router rip
 version 2
 redistribute ospf 1 metric 1
 passive-interface default
 no passive-interface GigabitEthernet0/1
 no passive-interface Serial0/0/1
 network 192.168.0.0
 no auto-summary
```

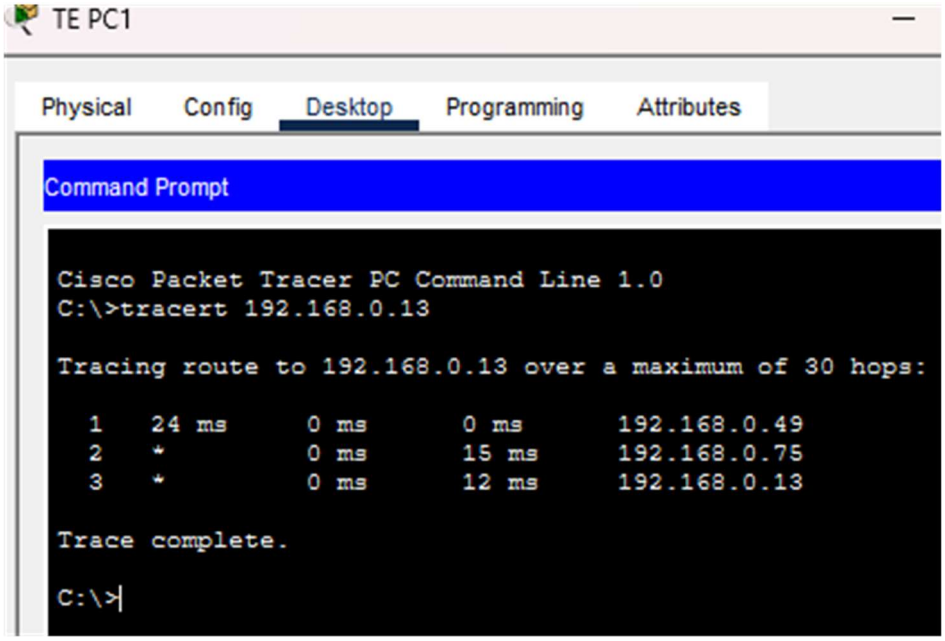
Failover Test Screenshots

Before Failover

1. Shared Router IP Route to DNS Server

```
Shared_Router# show ip route 192.168.0.13
Routing entry for 192.168.0.0/27
Known via "static", distance 1, metric 0
  Redistributing via rip, ospf 1
  Advertised by ospf 1 subnets
  Advertised by rip
  Routing Descriptor Blocks:
  * 192.168.0.75
      Route metric is 0, traffic share count is 1
```

2. Test Engineer PC Trace route to DNS Server



During Failover

1. Primary link shutdown to simulate a failover

```
Shared_Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Shared_Router(config)#interface gig0/1
Shared_Router(config-if)#shutdown

Shared_Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to
administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to down
end
```

2. Shared Router IP Route to DNS Server

```
Shared_Router#show ip route 192.168.0.13
Routing entry for 192.168.0.0/27
Known via "rip", distance 120, metric 1
  Redistributing via rip
  Last update from 192.168.0.114 on Serial0/0/1, 00:00:12 ago
  Routing Descriptor Blocks:
  * 192.168.0.114, from 192.168.0.114, 00:00:12 ago, via
Serial0/0/1
      Route metric is 1, traffic share count is 1
```

3. Test Engineer PC Trace route to DNS Server

```
C:\>tracert 192.168.0.13

Tracing route to 192.168.0.13 over a maximum of 30 hops:

  1    0 ms      0 ms      16 ms     192.168.0.49
  2    0 ms      1 ms      11 ms     192.168.0.114
  3    *         *         10 ms     192.168.0.13

Trace complete.
```

After Restoring

1. Primary link no shutdown to simulate a restoration

```
Shared_Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Shared_Router(config)#interface gig0/1
Shared_Router(config-if)#no shutdown

Shared_Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to up
end
```

2. Shared Router IP Route to DNS Server

```
Shared_Router#show ip route 192.168.0.13
Routing entry for 192.168.0.0/27
Known via "static", distance 1, metric 0
  Redistributing via rip, ospf 1
  Advertised by ospf 1 subnets
  Advertised by rip
  Routing Descriptor Blocks:
  * 192.168.0.75
      Route metric is 0, traffic share count is 1
```

Routing Information Protocol (RIP) is a distance vector routing protocol used to exchange data between routers. It measures distance based on hop count. It is usually used in small networks because it has a maximum hop count of 15 and slower convergence times compared to other protocols. Open Shortest Path First (OSPF) uses a link state routing algorithm to efficiently route traffic within a network (geeksforgeeks, 2025).

There are also a few technical issues which BrightSoft Solutions must address with the help of its network infrastructure. The first issue is the compatibility between the two protocols and the significant difference between the two protocols' metrics for determining the best route. For example, OSPF uses a complex cost calculation based on bandwidth, while RIP uses a simplistic 'hop count' approach. This leads to sub-optimal routing where the route is taken through a slow 10Mbps legacy network rather than a fast 1Gbps fiber network simply because the latter has fewer jumps (Cisco, 2018).

When routes are redistributed between the two protocols, the metric must be translated. For example, an OSPF route with a cost of 10 must be converted into a hop count when it is introduced into RIP. If the assigned metric is too low, it may lead to inefficient or suboptimal routing paths. If the metric is too high, the route may become unreachable because RIP has a maximum limit of 15 hops. Incorrect or inconsistent metric assignments can also create routing loops if they are not carefully configured (NetworkLessons, n.d).

The other major concern is the need to redistribute the route between these two protocols. For the Shared Router to share the route from the OSPF domain to the RIP domain, we need to manually redistribute this route, which can easily result in a routing loop (Harris, n.d). This, of course, directly impacts the effectiveness of our failover or backup link. If the primary OSPF link fails, the time that is taken to "converge" and provide a backup route is much longer using the RIP protocol due to the 30-second update timer, as opposed to almost instant OSPF "hello" packets. As a result, there is a great deal of potential downtime that is represented by this strategy, which could potentially drop sessions for the Developers or the Staging environment while the legacy protocol catches up. The backup link is designed to provide failover if the primary link fails. However, if the backup link uses RIP while the primary link uses OSPF (or the other way around), the failover process may not be smooth. This is because OSPF might still consider its route valid even when the physical link has failed, until its hello timers expire. At the same time, RIP may already have a route through the backup link, but since it has a higher Administrative Distance (AD), it will not be selected until the OSPF route is completely removed from the routing table (Cisco, 2024).

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